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# An open-source, 3D-printed auger powder doser designed with generative artificial intelligence†

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Many automated laboratories still weigh powders by hand, because commercial powder-dosing robots cost tens to hundreds of thousands of dollars. We describe a single-channel powder doser built from 3D-printed parts and off-the-shelf electronics for under $200, plus an analytical balance. A $6 microcontroller turns, tilts, and taps a printed auger, a tube with a spiral inside, until the balance reads the requested mass. The mechanical parts were first designed with generative artificial intelligence (AI) instead of conventional CAD software: the team made every design decision and reviewed, printed, and tested every part, while AI coding agents and a chat-driven CAD tool produced the geometry. Asking the AI for the whole assembly at once failed repeatedly; asking for one part at a time, against dimensioned drawings, worked better. The team later redrew the parts in conventional CAD; the tested doser uses some, including the auger and its drive gear. We tested 13 powders: seven food-safe and six relevant to alloy discovery. With fixed settings, ten powders moved 10–231 mg per auger revolution and three moved no measurable amount. In closed loop, 24 of 39 doses of 1 g landed within ±5% of the target, including every dose on the seven fastest-flowing powders dosed, and those on the three alloy-relevant powders within 1%. Doses of 50 and 200 mg were less reliable, because fast powders overshot and slow ones ran out of time; ongoing controller work targets this. The design files, firmware, data, and the prompts and responses exchanged with the AI tools are public at `https://github.com/vertical-cloud-lab/powder-doser`.

## 1 Introduction

Self-driving laboratories (SDLs) pair automated instruments with machine-learning planners that choose the next experiment, and they have sped up discovery in chemistry and materials science by an order of magnitude or more.[1–4] Software for running them,[5,6] methods for planning experiments,[7] and robots for handling liquids are now mature. Handling solids is not: most SDLs either work only with liquids or rely on a person to weigh out powders.[8,9] The gap matters most where the powders are the experiment. In alloy discovery by additive manufacturing (AM, the 3D printing of metal parts), each candidate composition is made by blending elemental or pre-alloyed powders,[10–14] so weighing many different powders accurately and repeatably is the step that limits the whole workflow.

Commercial powder-dosing systems exist and work well, but their cost and size put them out of reach of most academic laboratories. Benchtop gravimetric (weight-based) dispensers such as the METTLER TOLEDO XPR/CHRONECT line, Labman dosing stations, and MTI's AM powder dispensers cost from tens of thousands to several hundred thousand dollars. They are built around pharmaceutical and quality-control work and usually dose one vial at a time from proprietary dosing heads.[15,16] Pharmaceutical manufacturing meters powders at much higher flow rates with loss-in-weight feeders, which weigh the hopper continuously as it empties,[17–19] and laboratory studies have dosed milligram quantities with capillaries, vibrating chutes, and augers.[20–23] None of these is available as an open design that a laboratory can print, build, and modify. Open-source hardware has shown that 3D-printed instruments can reach research-grade performance at 1–10% of the commercial price and keep improving through community contributions,[24–27] and platforms such as Jubilee and low-cost SDL kits have brought that model to laboratory automation.[28,29]

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† Supplementary Information available: bill of materials; construction guide with an exploded view and illustrated assembly steps; auger and outlet geometry, with the tested and AI-modelled augers compared; the AI-assisted CAD workflow and a summary of the design log; balance noise at rest and after actuation; human tasks in the test campaign; the closed-loop dose record, with every attempt accounted for and example traces; composition error from dosing error; the test protocols as recorded, with run and record counts; run inventory; and AI usage log. See DOI: 00.0000/00000000.

This paper describes the first module of such an instrument: a single-channel powder doser in which a 3D-printed auger delivers powder into a cup on an analytical balance, and a microcontroller turns, tilts, and taps the auger until the balance reads the requested mass (Fig. 1). Each channel is a self-contained printable unit, and later versions will dose several powders (Section 4). We report the design, a test campaign on 13 powders with very different flow behaviour, and closed-loop dosing results, including where the current design fails.

This study's second contribution concerns how the doser was designed. Its parts were first designed without conventional CAD software such as SolidWorks or Fusion 360. They were modelled by generative artificial intelligence instead: large language model (LLM) coding agents wrote each part as code in the project's public repository, and we also explored a chat-driven CAD tool, Zoo Design Studio. The division of labour was the same throughout. People decided what to build, wrote the specifications, eventually supplied fully dimensioned drawings, and reviewed, printed, and tested every part. The AI tools turned those decisions into geometry. Research on generative CAD has produced large datasets and models,[30–33] and LLMs can now write CAD code (scripts that CAD software turns into solid models) from written descriptions,[34–37] but published evaluations test benchmark shapes rather than the design of a working instrument from start to finish. Every prompt, generated part, review comment, and failure in this project is preserved in a public repository,[38] so the doser doubles as a documented case study of AI-assisted hardware design. Once the AI-designed doser worked, the team recreated its parts in Fusion 360 to obtain editable models for release. Those sessions were recorded too, so they are documented in the same way and give a rough comparison between the two routes. The doser used for the tests in this paper carries some of these recreated parts, including the auger and the stepper pinion that drives it (see Experimental). We describe the workflow that worked best, the ways the AI tools failed, and what the approach cost and gained compared with conventional design.

## 2 Results and discussion

### 2.1 The dosing module

Fig. 1a,b shows the module as designed and as built. The printed auger tube is also the powder reservoir, so there is no separate hopper. Powder is poured in through the open upper end of the tube, which is then closed with a screw-on cap. The tube (25 mm outer diameter, 21 mm bore, 250 mm long) is printed as one piece. Along the third of its length nearest the outlet, a single-start screw (the flight, 10.4 mm pitch) winds around an 8 mm central core. The rest of the tube is plain and holds the powder (Fig. 1c). Two smaller augers, whose narrower inner channel is fed through a funnel at the top, hold about 3.3 and 9.2 mL, so the reservoir can be sized to the sample (SI Table S2). The whole piece turns together, like an Archimedes screw, rather than as a screw inside a fixed barrel. A NEMA-11 stepper motor turns it through a printed 20-tooth pinion and a 44-tooth gear printed on the outside of the tube, a 2.2:1 reduction. This auger and pinion are the Fusion 360 recreations described below. A push–pull solenoid taps the tube through a printed collar to help cohesive powders move. A small eccentric-rotating-mass (ERM) vibration motor was also planned, but its driver never worked (Section 2.4), so the doser as tested relies on rotation, tilt, and tapping. The tube rides on a hinged plate that two servos tilt through a 2:1 gear. The hinge axis runs just behind the outlet, about 12 mm from it in the current design, so the dose lands in nearly the same place at every angle (Fig. 1d): the tube can be parked horizontally, where nothing flows, and tipped up to dispense. The steepest tilt used in this work was 45°, which moves the outlet about 3 mm horizontally and 8 mm down. The servos can turn further, but steeper tilts were not tested.

An analytical balance (A&D HR-100A, 0.1 mg readability) weighs the collection cup and reports to a $6 Raspberry Pi Pico W microcontroller. The microcontroller runs a three-phase dosing procedure: fast continuous rotation, then small rotations, then solenoid taps until the reading is within tolerance of the requested mass (Fig. 1e; see Experimental). The structural parts are printed in polylactic acid (PLA), most of them with support material, and the remaining parts are commodity items. The bill of materials comes to under $200 for a single channel, excluding the balance, which most laboratories already own, and the custom circuit board, which a breadboard can replace (SI Table S1). That is roughly two orders of magnitude below a commercial gravimetric dosing head.

### 2.2 How the parts were designed

The mechanical design went through four versions in its first six weeks. Every step, then and since, is recorded in a design log in the repository, which now has 128 entries. The project began at a two-day hackathon in April 2026 as a hand-sketched mechanical scoop (a "powder excavator") meant to ride on an existing computer-controlled (CNC) gantry. An AI-assisted survey of small-scale dosing mechanisms[20,21] produced a list of eight candidates, among them a tap sieve, a metering wheel, a capillary wiper, and an auger. We chose the auger for its dose resolution, its printability, and its tolerance of different powders, in line with its wide use in commercial micro-feeders.[15,16] Before the second version, the team also printed four outlet geometries for the auger (SI Fig. S3). The second version packaged the auger, frame, drive, and actuators as a self-contained channel, the repeating unit of a future multi-powder doser. The third version kept the mechanism and changed how we asked for the parts. Instead of asking the AI tools for the whole module at once, we specified each part separately (auger, geared auger, bracket, tap collar, mounting plate, baseplate, and servo hinge). AI tools modelled each part, and we reviewed and printed it. The fourth version assembled these parts on the hinged baseplate with servo tilt. In June 2026, at our request, the coding agent also modelled a storage version of the auger, with the flight on only the third of the tube nearest the outlet, an open upper end, and a screw-on cap. Some parts changed when the team later redrew them in Fusion 360. The tested auger follows the storage design, with a 20:44 gear pair in place of the AI model's 16:48 and a uniform 0.5 mm flight at a slightly longer pitch (SI Table S3). The tap collar was also redesigned to hold a

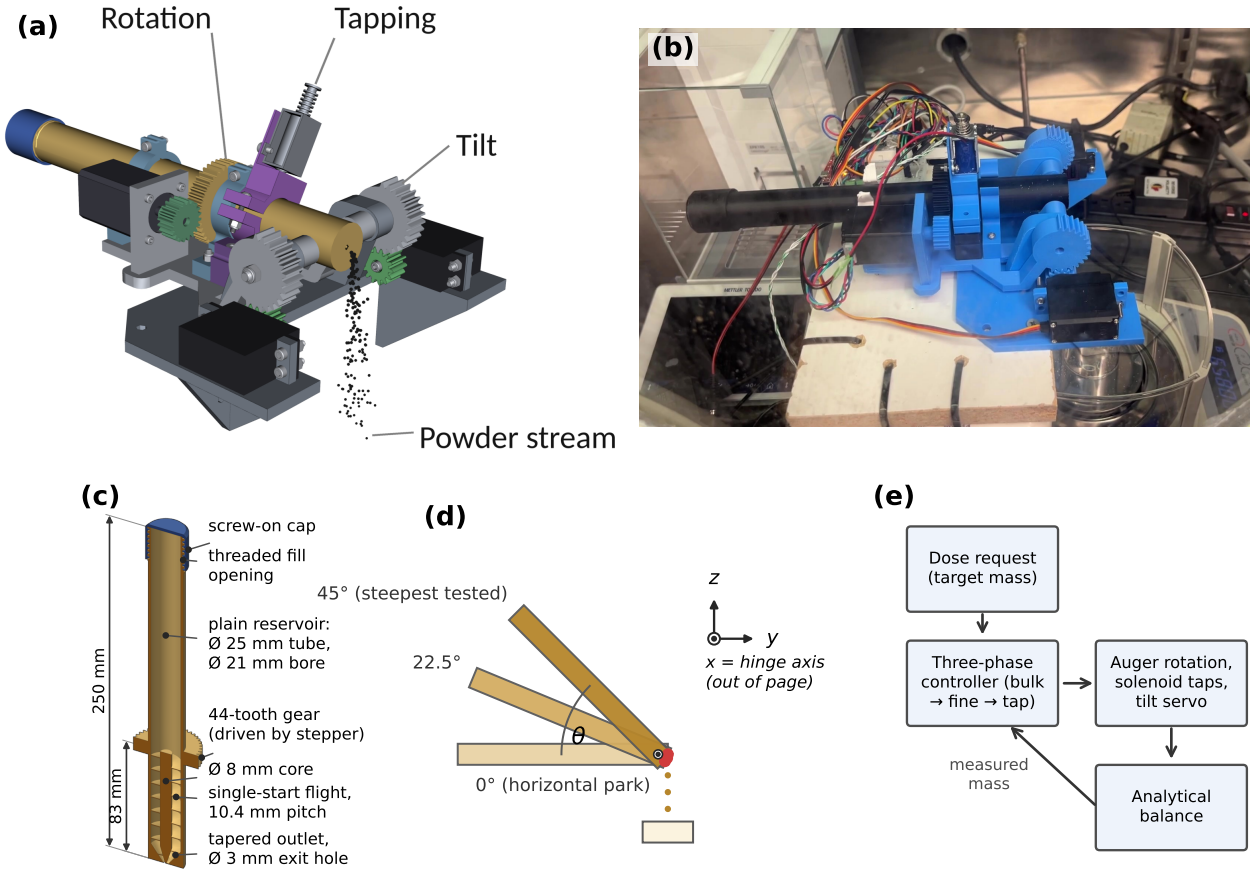



Fig. 1 The single-channel powder doser. (a) Annotated CAD model of the current design, showing the three motions used in this work: auger rotation, tapping, and tilt. Every printed part in this model is the team's Fusion 360 part except the tap-collar base, which is still the AI-modelled part. The stepper, servos, and solenoid are simplified models made from their datasheets (SI Figs. S1 and S2 show every part). (b) The module as built, photographed from the same viewpoint during its first automated dispense (sodium chloride into a crucible on the balance), inside a glove box at the University of Utah. The tests reported here were run on an open bench and in fume hoods. (c) Cut-away rendering of the tested auger and its screw-on cap, cut in half through the tube axis and drawn upright (outlet at the bottom). Cut faces are the flat, darker areas. The tube is itself the reservoir and is filled through the capped end. The flight and its core fill only the 83 mm nearest the outlet, and the 44-tooth gear is printed on the outside. Tube, flight, core, and gear are one printed piece and turn together. (d) Tilt about the hinge axis $x$, which runs about 12 mm behind the outlet (red), so the dose lands in nearly the same place as the tilt angle $\theta$ changes from 0° (horizontal park) to 45°, the steepest tilt tested. (e) Closed-loop dosing: the three-phase controller (bulk, then fine, then tap) compares the requested mass with the balance reading and drives the actuators until the two agree. Humans made every design decision and reviewed, printed, and tested every part. AI tools produced the original CAD models without conventional CAD software, and the team later recreated every printed part except the tap-collar base in Fusion 360.

different solenoid.

The specifications given to the AI grew steadily more detailed. The first requests described what the whole system had to do. Later requests described what each part had to do and how it met its neighbours: bolt patterns, shaft diameters, and mating faces. After that, the team supplied annotated engineering drawings, several of them fully dimensioned with the geometric relations between features. By the end, the AI's job had narrowed to modelling, that is, turning drawings into parametric CAD and working out derived dimensions and clearances. Every design decision stayed with the team.

Most parts were produced by LLM coding agents (the GitHub Copilot coding agent, mostly running Claude Opus models) working directly in the repository. Given a task written as a GitHub issue, the agent wrote the part as parametric code (Python with CadQuery, or OpenSCAD). In the same session it ran scripts that rendered views of the part, exported printable STL mesh files, and checked the geometry. The agent then opened a pull request, and we reviewed the part from the rendered images. SI Section S4 sets out this loop with an example drawing, the checklist of interfaces that requests specified, and when a part counted as finished. We also explored two commercial text-to-CAD tools on the same parts: CADSmith, and Zoo Design Studio, which pairs a code-based CAD engine, scripted in the KittyCAD Language (KCL), with a conversational design agent called Zookeeper. In Zoo Design Studio, too, parts were built mainly by chatting with the agent rather than by editing the model by hand. Every prompt, generated file, render, and review comment is preserved in the repository's pull requests and in a dedicated discussion thread.

#### 2.2.1 One part at a time

Early attempts asked the agent for the complete module from a description of its layout. The results looked plausible and did not work. The first tap collar merged several functions into one body, with overlapping features, wrong tolerances, and no room for the actuators it existed to carry (Fig. 2a). An early bracket arrived with its two mounting plates unconnected. Requests at the level of the assembly repeatedly produced overlapping solids, floating parts, and changes nobody had asked for, and fixing one defect often introduced another (Fig. 2d). The agent produced usable parts once each request covered a single part and listed where it connected to its neighbours (Fig. 2c). This fits the known weakness of current LLMs at reasoning about how shapes fit together in space.[34] Splitting a design into parts with defined interfaces is standard practice for human engineers, and when AI does the modelling it currently appears to be required. Table 1 lists the failure modes we met and what caught each one. Each design-log entry records what triggered the change, the reasoning, and the outcome (built and worked, built with issues, failed, or superseded). Of the 76 entries that made or changed a part model, the coding agent wrote 72 and 16 were printed. People reviewing renders caught 30 of the 41 recorded defects, printing or bench tests caught 6, and the agent's own checks caught 4 (SI Table S4).

Table 1 Failure modes of the AI modelling tools observed in this project, and what caught or prevented each one.

| Failure | Example | Caught or prevented by |
|---|---|---|
| Parts that do not fit together | Whole-module prompts; first tap collar; bracket with unconnected plates (Fig. 2a,d) | Asking for one part at a time, with interfaces listed |
| Unrequested changes | Fixing one defect altered an unrelated feature | Full re-inspection of rendered views at every iteration |
| Mislabelled variants | Two of four test nozzles swapped relative to their documentation | Human cross-check before printing |
| Sub-millimetre errors | Height mismatch of one print layer | Only the failed print |
| Designing around bad inputs | Out-of-date neighbouring parts led to a hole, platforms, then a gap (Fig. 2e–h) | Supplying the correct upstream files |
| No sense of "finished" | Late iterations disturbed features that were already right | The team freezing the geometry |
| Wrong orientation | Solenoid mis-oriented on the tap collar over several coding-agent iterations | Switching the part to Zoo Design Studio |

#### 2.2.2 Checking the output

Most of the effort went into checking. Because any iteration could change geometry outside the part being edited, every generated part had to be re-inspected in full. In one case the agent produced four miniature test nozzles and swapped the labels on two of them, which would have invalidated a dispensing comparison had a person not noticed. Automated checks (that a part is a single solid, that its surface has no holes, and that key interface dimensions are correct) caught a useful share of defects cheaply. Small errors still escaped both the checks and visual review: a height mismatch of one print layer was found only when the part peeled off the print bed. AI-generated alternatives were valuable early, while the design was still open, but late in the project the agent could not tell how close a part was to finished and kept changing features that were already right. AI-modelled parts therefore tended to stop at "good enough", because each further round of edits cost time and risked breaking something that worked.

#### 2.2.3 Comparing the tools

We used the coding agent from April to June 2026, CADSmith in May and June, and Zoo Design Studio from May to early July, mostly in June, so the comparison below describes the tools as they were then. The coding-agent workflow was fully traceable and needed nothing beyond a web browser, but each iteration took minutes and every change had to be reviewed. CADSmith, which we tried on the mounting plate and baseplate, returned only simplified, blocky geometry and was no more useful than the coding agent. Zoo Design Studio was the team's clear favourite. Its Zookeeper agent reasoned noticeably better about space, ran its own checks (such as counting solid bodies) before returning a result, and worked inside a live CAD view, so the team could inspect an editable model as it was built. Each iteration was slower,

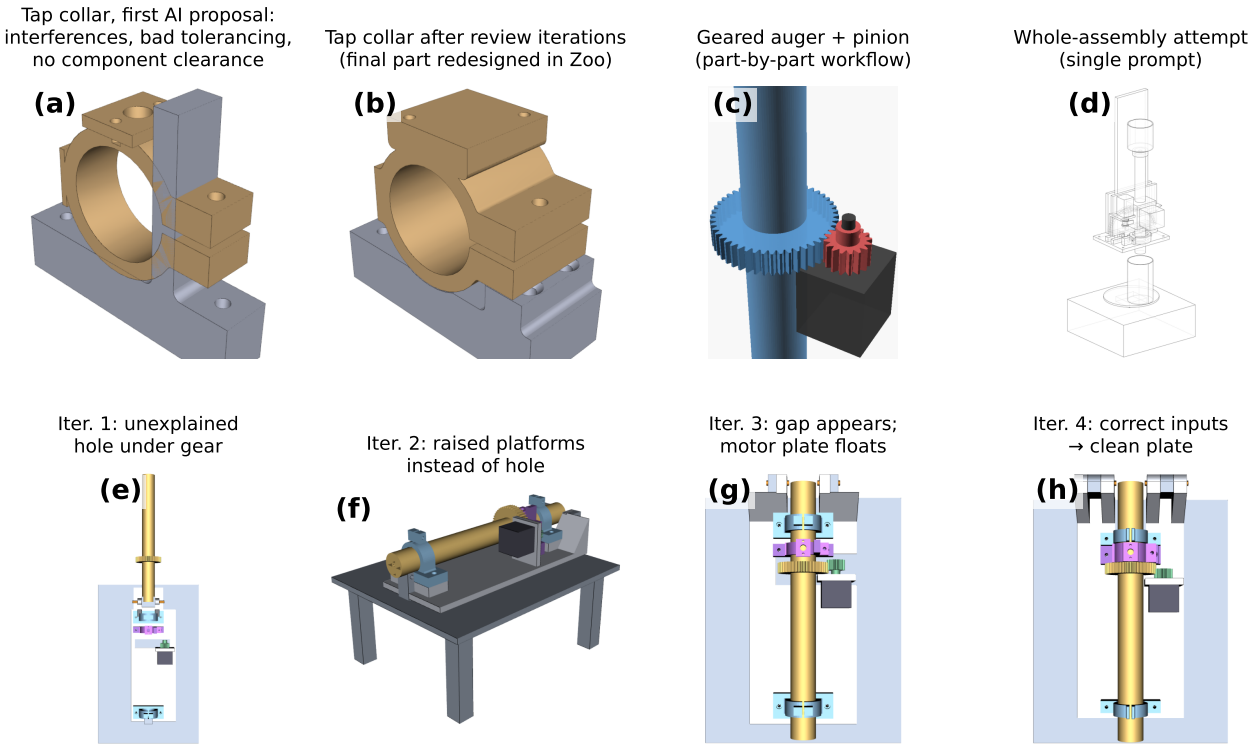



Fig. 2 AI-generated CAD, good and bad. (a) The first AI-generated tap collar: several functions merged into one body, with features overlapping one another, wrong tolerances, no room for the solenoid and vibration motor it had to carry, and no workable spatial layout. It could not have been built. (b) The same part after iterative review with the coding agent. The production collar was later remodelled in Zoo Design Studio, which reached a usable part in three iterations. (c) Geared auger and stepper pinion from the part-by-part workflow: the AI-modelled 16:48 pair, which the team later replaced with the 20:44 Fusion 360 parts used in the tests (Fig. 1c). (d) An attempt to generate the whole module from a single prompt: plausible at a glance, but with overlapping solids and floating parts that took longer to diagnose than the part-by-part route took to build. (e–h) Four review iterations of the mounting plate. The agent added an unexplained hole (e), then raised platforms (f), then a gap that left the motor plate floating (g), and produced a clean plate (h) only once the correct versions of the neighbouring parts were supplied. The cause was out-of-date input files, a human error that the agent silently designed around. All parts shown were modelled by AI tools to human-set requirements and reviewed by the team at each iteration. No conventional CAD software was used.

of the order of an hour, but on the parts we tried the team usually reached an acceptable design in fewer Zoo iterations, although we did not compare the two under controlled conditions. The final tap collar reached a usable design in three Zoo iterations after the same number of coding-agent iterations had failed to orient the solenoid correctly. At the time of testing, Zoo also had limits: imported reference parts could not be edited, there were no constraints between parts in an assembly, and manual edits were largely limited to changing parameter values, so the work stayed conversational. These rankings are the team's impressions from one project, and we did not measure them. A controlled comparison with identical prompts and parts is planned. Because these tools are changing quickly, the ranking may not last. In our more recent work, agents that drive a full CAD package through its application programming interface (API) have been among the most successful. An agent working only through Onshape's API assembled the current design from the team's part files, and the exploded view and build steps in SI Figs. S1 and S2 were rendered from that assembly. Improvements are visible across all of these tools, and open-source projects such as text-to-cad[39] are promising newer examples. We also judged that every AI route was slower, and gave lower-quality geometry, than experienced mechanical engineers working in conventional CAD software would have. The Fusion 360 recreations give a rough check. The narrated recordings for nine parts total about three hours of edited video,[40] roughly the length of a single Zoo Design Studio session. The recreations were not exact copies (the drive gears, for example, went from 16:48 to 20:44 teeth), and the recreated auger and pinion turned during every test reported here. The AI route did leave documentation behind as a by-product of the work. Every design intent, constraint, and revision exists as text, so the design can be reproduced, its history traced, and the project handed over more easily than with conventional CAD files alone.

### 2.3 How much powder one turn delivers

We tested the doser on 13 powders (Table 2). Seven are food-safe surrogates, chosen to cover a wide range of flow behaviour cheaply and safely: table salt (NaCl, the control), calcium lactate, xanthan gum, carboxymethyl cellulose (CMC), sodium alginate, and white and brown rice flour. The other six are relevant to the alloy-discovery work the doser is meant to serve. They are a gas-atomized AlSi10Mg alloy powder (near-spherical particles made by spraying molten metal), a coarse and a fine grade of silicon, sodium sulfate, barium chloride, and fumed silica, a very light flow aid. We ran each powder through the same fixed sequence of test protocols with frozen settings (Table 3), so that differences between powders reflect the powders rather than the settings. In the first test round (4–21 August 2026), the food-safe powders were tested on an open laboratory bench and the other powders in a shared fume hood. The balance became much noisier from 20 August: with nothing moving, eight successive readings varied by 0.0 mg on every run up to 12 August and by up to 88 mg afterwards (SI Fig. S5). Bench conditions are therefore confounded with powder group, because every research-relevant powder was measured in the noisier period.

With one set of settings, the mass delivered per auger revolution at a 22.5° tilt varied 24-fold across the ten powders that conveyed a measurable amount, from 10 mg for sodium alginate to 231 mg for AlSi10Mg (Fig. 3a). Seven powders (AlSi10Mg, coarse silicon, sodium sulfate, calcium lactate, barium chloride, xanthan gum, and salt) delivered 146–231 mg per revolution, and CMC, white rice flour, and sodium alginate delivered 10–26 mg. Three powders conveyed no reliably measurable amount at any tilt: brown rice flour, fine silicon, and fumed silica. The powder mattered far more than the settings. Splitting the variation in mass per revolution across all measured revolutions (on a log scale), 70% came from which powder was loaded, 20% from tilt, 4% from how each powder responded to tilt, and 6% from scatter between revolutions.

Delivery is not smooth. Balance readings streamed during continuous rotation rise in steps (Fig. 3b) that recur about every 4 s at 15 rpm, once per auger revolution. Across ten powders, the median period from autocorrelation was 3.94 s against a revolution time of 4.00 s. A single-start flight should behave this way, because each revolution carries one pocket of powder over the outlet. The pulsing limits how finely rotation alone can meter powder, and it is why the final approach to a requested mass uses taps (Section 2.5).

Plotting each powder's throughput against its variation from one revolution to the next (Fig. 3c) separates the powders into a readily dosed group, which delivers 150–230 mg per revolution with 2–18% variation, and a slow group that delivers 10–26 mg. Both numbers come from about two minutes of bench time per powder, which makes the plot a candidate for sorting new powders quickly. It is not yet a validated diagnostic, because we have not independently measured particle size, particle shape, or bulk density.

The two ends of the range are consistent with particle shape. Gas-atomized AlSi10Mg consists of near-spherical particles in a narrow size range, the properties laser powder-bed fusion requires, and it was the fastest powder tested. The fine silicon is milled, angular, and smaller than 44 $\mu$m (it passes a 325-mesh sieve). It did not move at all at 0° or 22.5°, and at 45° it produced a single 7 mg pulse in six revolutions. The coarse silicon, the same material at roughly 75–150 $\mu$m, delivered 211 mg per revolution. Forces that make particles stick together matter more relative to their weight as particles get smaller, which fits this sharp difference between the two silicon grades. We read the three powders that conveyed no measurable amount as a limit of this auger geometry rather than a fixed property of the powders. A wider flight, a larger pitch, or a taller fin would move that limit. The failures also differ in kind. Fine silicon moved slightly at the steepest tilt, so for it the driving force was too weak to overcome cohesion. Fumed silica, which is extremely light and fluffy, never delivered more than a few milligrams per revolution at any tilt and varied wildly from one revolution to the next, which points instead to trapped air.

Table 2 The 13 powders tested. Mass per auger revolution is the protocol C mean at 22.5° tilt (Fig. 3a). Closed-loop columns give the number of valid doses that finished within the acceptance limit (±10% at 50 mg, ±5% at 200 mg and 1 g) out of the number run (Fig. 5). A dash means not run. The silicon grades are named by sieve size: −325 mesh passes a 44 μm sieve, and −110/+200 mesh lies between about 75 and 150 μm.

| Powder | Group | mg per rev. | 50 mg | 200 mg | 1 g | Observed behaviour |
|---|---|---|---|---|---|---|
| AlSi10Mg (gas-atomized) | Research | 231 | 1/3 | 3/3 | 3/3 | Fastest powder; fluid-like; overshoots small targets |
| Silicon, −110/+200 mesh | Research | 211 | 2/3 | 2/3 | 3/3 | Free-flowing |
| Sodium sulfate | Research | 208 | 0/3 | 2/3 | 3/3 | Free-flowing; overshoots small targets |
| Calcium lactate | Surrogate | 198 | 1/3 | 2/3 | 3/3 | Free-flowing; largest tap increment |
| Barium chloride | Research | 185 | – | – | – | Absorbed moisture and caked in the tube in round 2 |
| Xanthan gum | Surrogate | 161 | 2/3 | 3/3 | 3/3 | Flows under rotation; can clog after bursts |
| Sodium chloride (control) | Surrogate | 146 | 2/3 | 2/3 | 6/6 | Free-flowing baseline |
| Carboxymethyl cellulose | Surrogate | 26 | 3/3 | 1/3 | 3/3 | Slow; charges electrostatically |
| White rice flour | Surrogate | 13 | 3/3 | 3/3 | 0/3 | Slow and pulsing; most variable between revolutions |
| Sodium alginate | Surrogate | 10 | 0/3 | 0/3 | 0/3 | Slowest powder that moved |
| Silicon, −325 mesh | Research | none | 0/3 | 0/3 | 0/6 | Did not move at 0 or 22.5°; one pulse at 45° |
| Brown rice flour | Surrogate | none | – | – | 0/3 | Did not move at any tilt |
| Fumed silica | Research | none | – | – | – | Did not move; very light and fluffy |

Table 3 Test protocols. Every powder went through the same protocols in the same order with frozen settings, so results can be compared across powders. The sequence characterizes the powders and does not tune anything. Rotations and speeds refer to the auger, and tilts are the tube's angle above horizontal, from 0° (horizontal park) to 45° (the steepest used). A trial is one recorded measurement. SI Table S10 gives the settings as the raw records store them and the number of records each protocol produced.

| Protocol | Varied | Fixed settings | What it measures | Trials |
|---|---|---|---|---|
| A Balance baseline | Nothing | Tilt 22.5°; eight readings, no actuation | Noise floor of the run, against which every other protocol is read | 8 |
| B Static hold | Tilt (0, 22.5, 45°) | 15 s hold at each tilt, no actuation | Whether any powder flows on its own (clean shut-off) | 3 |
| C Rotation | Tilt (0, 22.5, 45°) | Six single revolutions per tilt at 30 rpm, with a settled reading after each | Mass per revolution and its revolution-to-revolution variation | 18 |
| D Speed sweep | Speed (15, 45, 90 rpm) | Tilt 22.5°; three revolutions commanded per speed; balance streamed every 0.25 s | Delivery rate against speed; mass-against-time traces | 3 |
| E Tap | Tilt (0, 22.5°) | Eight trials of a one-revolution re-feed followed by one 60 ms solenoid tap | Mass released by a single tap, with the re-feed logged separately | 32 |
| F Vibration | Tilt (0, 22.5°) | As E, with three vibration bursts in place of the tap | Mass released by vibration (no data: driver fault) | 0 |
| G Closed-loop, 1 g | Nothing | Three doses of 1.000 g; three-phase controller | Dose error, time, and effort per phase | 3 doses |
| H Closed-loop, small | Target (50, 200 mg) | Three doses at each target; same controller | Dose error and time at small targets | 6 doses |

### 2.4 Tilt and tapping

Tilt is the strongest setting the controller has. For the nine powders measured at all three angles, raising the tube from horizontal to 22.5° increased delivery by a median factor of 4.2, and raising it further to 45° added a median factor of 1.2 (Fig. 4a). The cohesive powders depended most on gravity: sodium alginate delivered 15 times more per revolution at 45° than horizontally and white rice flour 10 times more, against about 5 times for calcium lactate and sodium sulfate. Tilt was also the only setting that moved one of the non-conveying powders at all (fine silicon, at 45°).

With the auger stopped, nothing flowed on its own. In 33 fifteen-second holds, covering all three tilts on the 11 runs recorded up to 12 August, the balance reading changed by 0.0 mg every time (protocol B). From 20 August, hold readings scattered by tens of milligrams in both directions, including at 0° where gravity cannot empty the tube, so they reflect the bench rather than the powder. Parking the tube horizontally therefore gives a clean shut-off, at least to the resolution of the quiet-bench runs.

A single solenoid tap released between 0.2 mg (sodium alginate) and 20 mg (calcium lactate), or 0.6–10% of one revolution's delivery (Fig. 4b). The size of a tap does not simply follow throughput. Barium chloride and calcium lactate delivered similar amounts per revolution (185 and 198 mg) but released 2.1 and 20 mg per tap, a tenfold difference. Tap increments could be resolved only on the quieter runs. On most fume-hood runs they were smaller than the bench noise.

The vibration motor could not be characterized: its driver (DRV2605L) returned an input/output error in every run that asked for protocol F. The speed sweep (protocol D) suggests that delivery rate grows more slowly than speed, with a median exponent of 0.78 across ten powders for rate $\propto$ speed$^{\alpha}$ over 15–90 rpm (above 1 only for white rice flour, CMC, and sodium alginate). Each speed was measured once, in a fixed order, without re-zeroing the balance, and a firmware timing error let the auger turn 15–54% further than commanded, so we treat these exponents as an indication only and do not plot them.

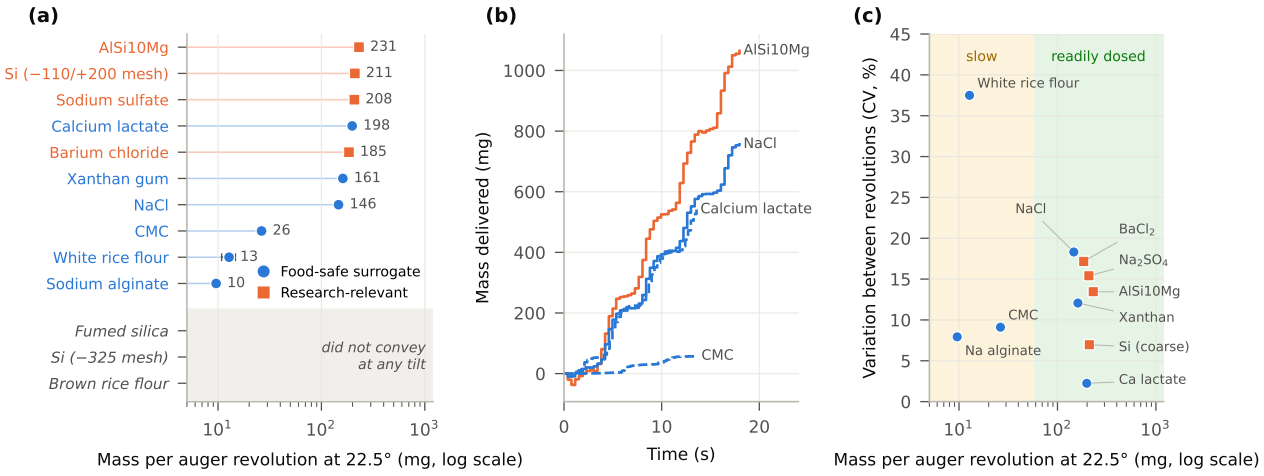



Fig. 3 How much powder the auger delivers, for 13 powders under one fixed set of settings. (a) Mass delivered per auger revolution at 22.5° tilt (protocol C, six revolutions at 30 rpm, with standard-error bars). The three powders at the bottom delivered nothing distinguishable from zero at any tilt and are listed without a value. (b) Balance readings streamed while the auger turned continuously at 15 rpm and 22.5° tilt (protocol D), for four powders. The mass rises in steps about 4 s apart, one per auger revolution. (c) Mass per revolution against the variation between successive revolutions (coefficient of variation of six revolutions). Both numbers come from protocol C alone. The shading marks the two groups discussed in the text. Blue circles are food-safe surrogates and orange squares research-relevant powders. One run per powder is shown (see Experimental). Fill level was not controlled, particle size and shape were not measured, and all research-relevant powders were tested in the noisier period. Rotations and speeds refer to the auger, and angles to the tube's tilt above horizontal.

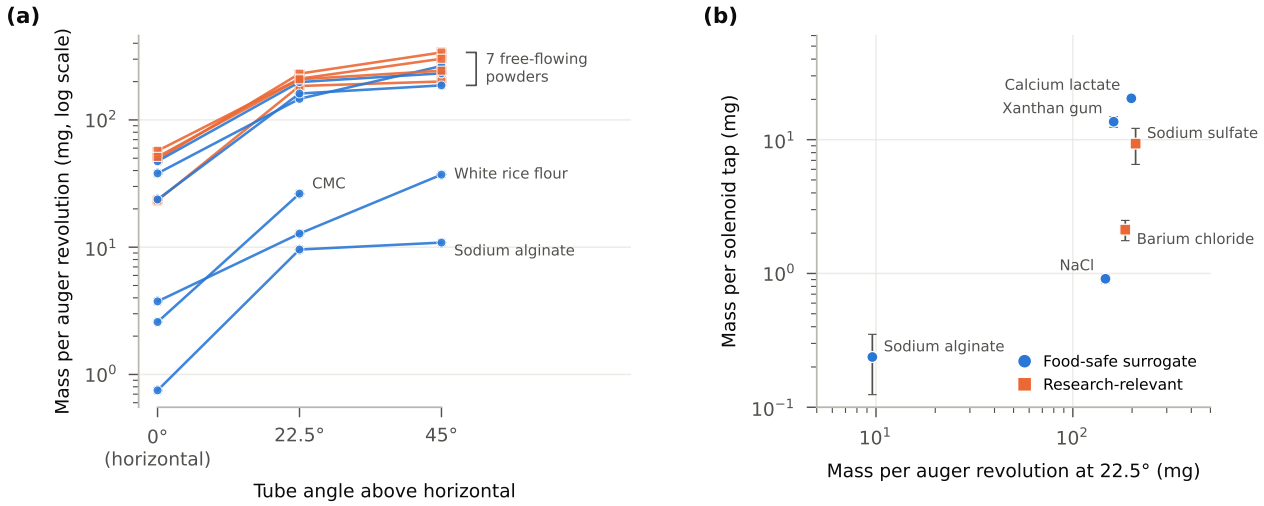



Fig. 4 Effect of tilt and tapping. (a) Mass per auger revolution with the tube horizontal and tilted to 22.5° and 45° (protocol C) for the ten powders that conveyed a measurable amount. A point is shown only when its mean is more than twice its standard error, that is, clearly larger than the scatter. CMC at 45° did not meet this rule. The seven free-flowing powders overlap at the top and are bracketed, and Table 2 gives their values. On the logarithmic axis, equal vertical distances are equal ratios. (b) Mass released by one solenoid tap (protocol E, mean and standard error of eight taps at 22.5°) against mass per auger revolution, for the six powders whose tap increment exceeded twice its standard error. Markers as in Fig. 3.

### 2.5 Closed-loop dosing

The controller doses in three phases: continuous rotation until the cup is within 0.5 g of the requested mass, then 45° rotations with a settled balance reading after each, and finally pairs of solenoid taps until the reading is within ±5 mg (see Experimental). All doses used one set of phase settings, tuned by hand on salt and frozen for every powder. In the first test round we ran three doses of 1 g (protocol G) on each powder whose bench conditions allowed it. A second round, from 3 to 15 September 2026, added three doses each of 50 mg and 200 mg (protocol H) and the 1 g doses that bench noise had prevented for the research-relevant powders. Doses are scored against acceptance limits fixed before testing: ±10% of the requested mass below 100 mg and ±5% at or above it. If dose errors are spread evenly within these limits, the composition error of an equimolar five-component blend[11,12] is 0.5 atomic percent (one standard deviation), with a worst case of 1.6 atomic percent. The aluminium-rich blends planned for our alloy work, in batches of 10–500 g, stay near 0.4 atomic percent even in the worst case (SI Section S8). Each delivered mass is also recorded, so the as-dosed composition is known to within 0.1 atomic percent. Processing can shift composition far more: directed energy deposition from elemental powders has shifted it by 5–10 atomic percent.[12,41] Failed doses stay in every count, and SI Table S7 accounts for every attempt, including those excluded and the combinations never run.

At 1 g, 24 of the 39 valid doses finished within the ±5% limit (Fig. 5c). They were every dose on the seven free-flowing and moderately slow powders: salt, calcium lactate, xanthan gum, CMC, AlSi10Mg, coarse silicon, and sodium sulfate. The three research-relevant powders were the most accurate: all nine of their 1 g doses finished within 10 mg (1%) of the target, with a median time of 111 s. The four powders that failed at 1 g failed for two different reasons. Fine silicon and brown rice flour delivered almost nothing, as in the open-loop tests. White rice flour and sodium alginate did move, but too slowly: they used up the controller's budget of 200 small rotations and stopped 14% and 29% short of the target after about 14 minutes.

Smaller doses were harder, in different ways for fast and slow powders (Fig. 5a,b). At 200 mg, 18 of 30 doses finished within ±5%, including all three for AlSi10Mg, xanthan gum, and white rice flour. At 50 mg, 14 of 30 finished within ±10%, although most of these passes lie within the balance noise of the limit (Section 2.7). The fast powders now overshot: of the 14 doses on free-flowing powders that missed the limit at 50 or 200 mg, 13 were overshoots, including all three sodium sulfate doses at 50 mg. For these powders a single 45° turn of the auger delivered a median of 30 mg, against 2 mg for the slow powders, and delivery arrives in pulses (Fig. 3b). The last step of a 50 mg dose on a free-flowing powder therefore added a median of 46 mg, enough to step over the whole acceptance band. The controller also had a flaw at small targets. It was meant to skip continuous rotation, but because the threshold for that phase equalled the target, a tared reading a fraction of a milligram below zero started a brief continuous spin. This happened in 19 of the 60 valid 50 and 200 mg doses and delivered up to 103 mg (SI Fig. S7 shows representative traces). Slow powders undershot instead, for the same reason as at 1 g. Overshoot cannot be undone in powder dosing, so for small targets a finer final increment matters more than a faster one.

Time was the main cost of accuracy (Fig. 5d). Doses that finished within the limit took a median of 53 s at 50 mg, 152 s at 200 mg and 185 s at 1 g, far longer than the 30 s we had aimed for. Slow powders took much longer: white rice flour needed 11–13 minutes for each 200 mg dose. The fastest "doses" were failures: fine silicon and brown rice flour stopped after 6–52 s, when the controller detected that nothing was arriving.

Two features of the controller explain most of the remaining failures. First, it never returns to an earlier phase. Once the tap phase begins, it cannot rotate the auger to refill the outlet, apart from 5° nudges that proved too small for the slower powders. When the flight nearest the outlet runs empty, taps alone cannot finish the dose. This is how the first-round 1 g doses of calcium lactate, xanthan gum, and CMC ended: each came to rest 15–49 mg short after 66–166 taps, inside the acceptance limit but outside the controller's own stopping band. Second, that stopping band was ±5 mg at every target. At 1 g it is ten times tighter than the ±50 mg acceptance limit, so the firmware logged these doses as stalled although they met the specification. We report every dose against the acceptance limit and keep the firmware's own label in the per-dose record (`paper/figures/data/doses_all.csv`, summarized in SI Table S8). A controller that can return to rotation when taps stop working, and whose stopping band scales with the target, is the most direct improvement.

### 2.6 A first trial in a glove box

Before the test campaign, on 24 July 2026, the doser had a short qualitative trial inside an inert-atmosphere glove box in a polymer chemistry laboratory at the University of Utah (Fig. 1b). Team members typed requests to an AI agent in a GitHub issue, and the agent sent the commands to the doser over the network, so nobody had to reach into the glove box to run it. The first 1 g dose of salt needed two runs of the controller, because salt fed slowly with the settings then in use, and it ended 4.4 mg short. In the smallest auger, chunks of a solid norbornene monomer dispensed well with the three-phase procedure. A mid-sized auger loaded with about 2 g of fine, hair-like azobisisobutyronitrile (AIBN, a polymerization initiator) appeared to clog and delivered nothing. Recrystallizing AIBN into coarser grains may be a simpler fix than changing the doser. Two problems were specific to the glove box. Static charge in the dry atmosphere made salt cling to the inside of the auger, and vibration from the glove box and the motor slowly loosened the nuts until the gears no longer meshed. Locknuts, and brackets printed in one piece with the plate, address the loosening. An ionizing fan, which was on hand but not used, may help with the static. The chemists who hosted the trial asked to build several dosers of their own.

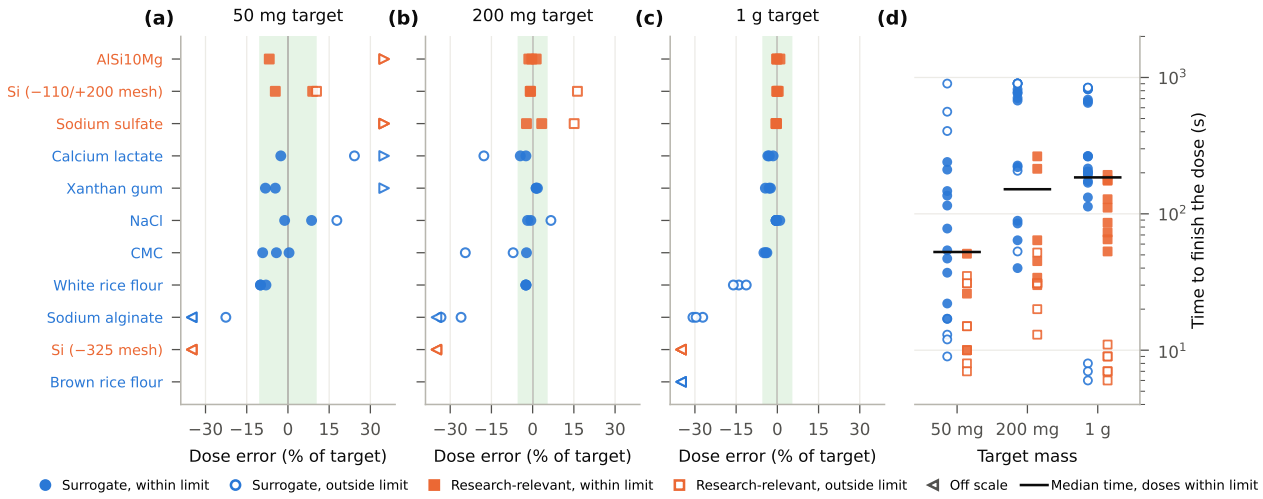



Fig. 5 Closed-loop dosing with one frozen set of controller settings. (a–c) Error of each valid dose at 50 mg, 200 mg and 1 g targets, as a percentage of the target. Green bands mark the acceptance limits (±10% below 100 mg, ±5% at or above). Filled markers finished within the limit and open markers outside it. Triangles at the edge are off scale (the powders that delivered little or nothing). Replicates share a row and are not jittered. Powders are ordered by mass per revolution (Fig. 3a). Doses at 1 g for the food-safe powders are from the first test round (protocol G, August 2026). All 50 and 200 mg doses and the 1 g doses for AlSi10Mg, coarse silicon, and sodium sulfate are from the second round (protocols G and H, September 2026). Barium chloride has no valid closed-loop dose (it caked in the tube) and fumed silica was not dosed. (d) Time to finish each dose, with horizontal bars at the median of the doses within the limit.

### 2.7 Limits of the current design

Several limits follow from these results. First, the design covers doses from tens of milligrams to grams. Doses of 0.1–10 mg, which some chemistry applications need, are out of reach for now. They would require per-powder calibration, better isolation from vibration, and a smaller auger. Second, three of the 13 powders conveyed no measurable amount, so the current geometry sets a floor on how cohesive or how light a powder can be. Third, the outcome of a dose depends on how much powder happens to sit in the screw near the outlet when the dose starts. There is no defined starting state, and in routine use the auger would never be reset between doses anyway, so this variability is part of the real performance. Fourth, all masses were weighed in the cup, so powder that left the tube but missed the cup is not counted, and the tube's fill level was not controlled. Fifth, the doser ran each set of doses unattended, but people filled and mounted the augers, emptied the cup between runs, and recovered from faults, most of which involved the balance or its surroundings (SI Table S6). Finally, the balance sets the noise floor. Idle readings drifted by up to tens of milligrams in late August. In the second round, readings at rest scattered by less than 0.4 mg, but readings taken right after the doser moved scattered by 5–19 mg on half of the runs with small doses (SI Fig. S6). Only one of the 14 passing 50 mg doses lay more than twice that noise inside its limit, so the 50 mg pass rate is uncertain, while the failures are clear. Isolating the balance from the doser's motion is therefore as important as refining the controller.

### 2.8 Cost and accessibility

The single-channel module (printed parts, motors, drivers, and microcontroller) costs under $200 in single quantities, excluding the balance (SI Table S1), and prints in under 24 h on a consumer printer. Commercial automated gravimetric dosing modules are quoted at $30k–$300k depending on configuration. Even where commercial instruments are more accurate, a cost gap of two orders of magnitude changes what a laboratory can do. It can dedicate one printed channel to each powder so that powders never mix, throw away and reprint the parts the powder touches, and replace a failed channel within a day. This is the economic argument for open-source laboratory hardware[24,25] applied to the solid-handling gap in SDLs.

## 3 Experimental

### 3.1 Fabrication

We printed all parts in PLA on a Bambu Lab H2D printer (0.2 mm layers, four perimeters, 25% infill), most of them with support material. The auger used in every test is a single printed piece modelled by the team in Fusion 360 (Fig. 1c). It is a tube 250 mm long, with a 25 mm outer diameter and a 21 mm bore. An 8 mm central core and a single-start flight, 0.5 mm thick with a 10.4 mm pitch, run along the 83 mm nearest the outlet, which tapers to a 3 mm exit hole. A 44-tooth gear (module 1) is printed on the outside of the tube 78–88 mm from the outlet, and an external thread at the upper end carries the screw-on cap through which the tube is filled. We measured these dimensions from the part file. A 20-tooth pinion on the stepper drives the 44-tooth gear (2.2:1 reduction), and the servos drive the tilt plate through a 2:1 gear. The auger, its cap, and the stepper pinion were each modelled in a narrated recording (`https://youtu.be/sKrvGBTUc8U`, `https://youtu.be/EID3ppxmE1U`, and `https://youtu.be/EO9qnYssKRQ`), as described in repository issues #117 and #120. The first AI-modelled geared auger (May 2026) instead had a flight along its

whole length, four loading slots in its closed top end, and a 16:48 gear pair (3:1). The agent's later storage version already had the capped upper end and the flight confined to the outlet third, so the team's main changes were the gear pair, the flight, and the cap thread. SI Table S3 compares the augers feature by feature. One AI-modelled part is still on the doser: the tap-collar base, a small plate under the tap collar whose raised stop keeps the collar from turning with the auger. The coding agent wrote it as parametric Python code in May 2026 (repository pull request #51), and SI Table S2 links its file. The AI-modelled auger, gears, bracket, tap collar, mounting plate, hinged baseplate, and servo gear sector are parametric models; their source files (CadQuery, OpenSCAD, and KCL), printable STL files, and renders for every iteration are in the repository. Purchased parts: a NEMA-11 stepper motor (auger drive), two MG996R-class servos (tilt), a 12 V push–pull solenoid (tapping), a coin-type ERM motor (vibration; unused, see Section 2.4), and an A&D HR-100A analytical balance (0.1 mg readability). The auger turns in printed collars, without bearings.

### 3.2 Electronics and firmware

A Raspberry Pi Pico W runs MicroPython firmware with commands to rotate by a given angle, tap, vibrate, tilt to an angle, and read the mass. A Pololu Tic T500 drives the stepper over a serial (UART) link, a DRV8871 H-bridge drives the solenoid, and a DRV2605L haptic driver runs the vibration motor over $I^2C$. The balance is read over RS-232 through an SP3232 transceiver board. The electronics began on a breadboard and now sit on a custom printed circuit board (PCB). Schematics, board files, the wiring guide, and the firmware are in the repository (`hardware/`).

The dosing firmware includes both gear ratios, so its rotation commands and records are in auger revolutions, auger rpm, and tube degrees. The one exception is the test battery, whose raw records store the tilt as a servo setting rather than an angle. The 2:1 tilt gear halves that setting, so the battery's three tilts are tube angles of 0°, 22.5°, and 45°. We report tube angles throughout and convert the raw settings once, in the analysis scripts. SI Section S9 maps the raw settings to angles. The auger ratio is confirmed independently by the protocol-D traces, whose pulses recur once per commanded auger revolution (3.94 against 4.00 s).

### 3.3 Powders and test protocols

Table 2 lists the 13 powders. Each powder had its own printed auger and cap, labelled with the powder's name, so no auger carried two powders. We filled each auger once through its capped upper end, without controlling the fill level or weighing it, and stored it loaded with tape over the outlet. A run began by mounting the filled auger and removing the tape, with occasional top-ups. Within a run nobody acted between doses: each dose was weighed from its own tare with earlier powder left in the cup, and the cup was emptied between runs. SI Table S6 lists every task a person did in the campaign. The test protocols (Table 3) ran in a fixed order with frozen settings, identically for every powder. Each protocol emitted machine-readable trial records over the serial link, and we archived them as versioned CSV files stamped with the powder, firmware version, and settings, so every number in this paper can be traced to its raw balance record. A trial is one recorded measurement: in protocol C one trial is a single auger revolution, so its 18 trials are six revolutions at each of three tilts. Protocol D streamed the balance every 0.25 s nominally (0.29–0.39 s in practice) while the auger turned continuously. Its timing loop advanced by the nominal poll interval rather than the elapsed time, so the auger turned 15–54% further than the three revolutions commanded per speed. Masses and rates from protocol D use the elapsed time.

We screened each run for quality before analysis. For example, we excluded a run with a failed tilt servo, a deliberate stress test of the bench, and runs in which the balance could not be read. One run per powder is shown in Figs. 3 and 4: the newest run that passed screening, preferring runs whose protocols C and E agreed. We report a mean as a measurement only if it exceeds twice its standard error, and otherwise report the powder as not having moved. Fumed silica is the one exception to screening. Its only run failed screening because the outlet could not be seen on camera to rule out a blockage, so its result (an average of at most 0.25 mg per revolution over 350 revolutions) is reported but is less certain than the others. The noise floor of each run is the spread of its eight protocol-A readings.

### 3.4 Closed-loop dosing procedure

The controller has three phases, with settings that were the same for every powder and every dose. In the bulk phase it tilts the tube to 45° and turns the auger continuously at 55 rpm, stopping 0.12 g early to allow for powder still in flight, until less than 0.5 g remains. In the fine phase it tilts the tube to 22.5° and turns the auger in 45° steps, waiting 1.5 s for a settled reading after each, until less than 50 mg remains. In the tap phase it lowers the tube to horizontal and fires two 60 ms solenoid taps per cycle, again reading the balance after each cycle, until the reading is within ±5 mg of the target. If three tap cycles in a row add no mass, it turns the auger forward 5° to refill the outlet, up to ten times. For the smaller protocol-H doses, the hand-over points were capped at the target mass (bulk) and at half the target (fine). At 50 mg, for example, the controller is meant to skip fast rotation and to switch from small rotations to tapping when 25 mg remains. A tared reading slightly below zero could still start a brief continuous spin, because the bulk threshold equalled the target (Section 2.5). A dose ends when the reading is within the ±5 mg band ("ok"), when it exceeds the target by more than the band (overshoot), when tapping and nudging add nothing more (stalled), after 200 cycles of one phase (cycle budget), or after 900 s. We tuned the settings by hand on salt and then froze them, and no dose outcome was used to retune them. In the first round the controller used a reading only once the balance flagged it as stable. After the move to a second fume hood, the balance rarely reported stability while the rig was running, so the second round averaged a bracket of five readings taken 0.25 s apart instead. Before each set of doses, the balance had to stay within the dose tolerance while nothing moved. Runs that could not pass this check were postponed. Our reporting follows

the gravimetric structure that ISO 8655-6 prescribes for piston pipettes, as adapted for an open-hardware pipette:[42,43] replicate doses at several target masses, each summarized by its systematic error (mean deviation from target), random error (standard deviation), overshoot rate, and time. Every dose, including failures and the firmware's own termination label, is listed in the per-dose record in the repository (`paper/figures/data/doses_all.csv`) and summarized by powder and target in SI Table S8.

### 3.5 Safety

Several of the powders are hazardous, and anyone repeating this work should handle them as their safety data sheets direct. Fine metal and semimetal powders, such as AlSi10Mg and the −325 mesh silicon, are combustible dusts: keep the quantities small, avoid raising dust clouds, and keep them away from sparks. The stepper motor, the solenoid, and static charge on the printed PLA parts are possible ignition sources. Barium chloride is toxic if swallowed and harmful if inhaled. Fumed silica, fine silicon, and the flours form fine dusts that should not be breathed in, so handle them in a fume hood or another ventilated enclosure. The AIBN used in the glove-box trial can catch fire if heated. The gears and the tilt hinge are pinch points while the doser is moving.

### 3.6 AI design workflow and logging

LLM coding agents carried out tasks written as GitHub issues: the GitHub Copilot coding agent from April to June 2026, when the AI-modelled parts were made, and Anthropic's Claude Code agent from the end of June 2026. Where the records name the model, the Copilot agent mostly ran Claude Opus 4.7 or 4.8, and Claude Code ran Claude Opus 4.8, Opus 5, Opus 5.5, Fable 5, and Fable 5.1 (SI Table S12). Each design session produced a pull request containing the generated CAD code, rendered orthographic and isometric views, and STL files, which the team reviewed in the browser. We ran literature and design-review queries through Edison Scientific. We explored CADSmith (May and June 2026) and Zoo Design Studio (mainly June 2026, after trials of Zoo's text-to-CAD service from May) on the same part specifications. In Zoo Design Studio, modelling was done through its Zookeeper agent. No conventional CAD software was used in first designing the parts. The Fusion 360 recreations described in the Results came afterwards, and the tested rig includes some of them (see Fabrication). AI tools also helped prepare this paper. Claude Code wrote the analysis and figure scripts and drafted and revised the manuscript and SI. It worked from team members' requests and review comments on the manuscript pull request (#97) and linked issues. Edison Scientific ran literature searches and mock reviews of drafts. References were checked against their DOI records where these exist (`paper/references_validation.md`). In line with *Digital Discovery*'s guidance on the use of LLMs, the prompts and responses (issues, pull-request threads, and Zoo Design Studio transcripts) are public in the repository, with the date of every exchange, and are summarized in the SI. The raw agent session logs kept by GitHub Actions expire after 90 days. Those from April to early July had expired by October 2026; the later logs and the Claude Code session transcripts were downloaded then and are part of the archived release. Over the project, team members addressed the two coding agents 754 times, and the agents authored 90% of all commits, including those for firmware, analysis, and writing (SI Fig. S8).

## 4 Conclusions

A single-channel powder doser with closed-loop weighing can be built from printed parts and commodity electronics for under $200 plus a balance. With one fixed set of settings, it delivered 10–231 mg per auger revolution for ten of 13 powders and no measurable amount of the other three. In closed loop, 24 of 39 doses of 1 g finished within ±5% of the target, including every dose on the seven fastest-flowing powders dosed. Doses of 50 and 200 mg met their limits in only 14 and 18 of 30 doses, and passing doses took a median of about 1–3 minutes. Its main limits are an auger geometry that cannot move very fine or very light powders, a controller that cannot recover once the outlet runs empty, and a balance whose noise sets the floor for small doses.

The module is already in use beyond the tests reported here. In July 2026 it dosed salt and a solid monomer inside a glove box, run by an AI agent over the network (Section 2.6). In late September, with settings from the controller optimization described below, it weighed about 8 g of AlSi10Mg into the cup of a custom charge, which the lab's ultrasonic atomizer then turned into new powder. It also dispensed 8 g of an Al 4047 powder that the same atomizer had made.

The design process is the second result, although it comes from one team and one project. Generating a whole assembly with AI did not work. Generating one part at a time against dimensioned drawings, with automated checks and human review, worked better, and it left behind a complete written record of every decision. The 128-entry design log, failures included, is itself a dataset for studying AI-assisted hardware design.

This module is the first of several planned stages. The first extension is a multi-powder doser. An early concept arranged several printed channels around a shared cup. The current design instead carries passive auger cartridges on a roller chain past a single fixed station that turns, taps, and weighs them, and a first printed prototype exists (Fig. 6). The team modelled its parts in Onshape, a conventional CAD package. The second is automatic calibration for each powder. This paper sets the phase thresholds, tilt angles, speeds, and tap schedules by hand. Bayesian optimization could tune them for each powder, subject to the constraint that an overshoot cannot be undone, and a formulation of that problem is already in the repository. Work on optimizing the controller is under way in the lab, and better control should make small doses of 50 and 200 mg more reliable. The third is a public database of tuned settings for many powders, linked to measured powder properties. The test protocols may help build it. The two numbers in Fig. 3c come from protocol C alone, which takes about 2 min per powder. If they track independent measurements such as bulk and tapped density, the doser could classify a new powder as a side effect of dosing it, although that idea is still untested. We also plan experiments on the mechanism itself, including pausing the auger part-way through a revolution and printing transparent

augers to watch the flight fill and empty, and changes to the auger geometry aimed at the powders that did not convey. Finally, we will keep evaluating generative design tools, including AI tools for laying out printed circuit boards like the one that now carries the doser's electronics.

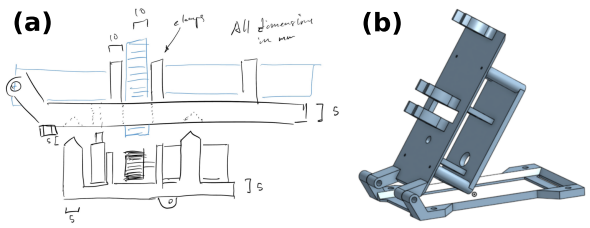



Fig. 6 Future work: the multi-powder doser, in which passive auger cartridges ride on a roller chain past one fixed station that turns, taps, and weighs them (repository issue #128). (a) Hand sketch of the auger carriage, drawn before modelling (dimensions in mm). (b) The auger and frame carriages of the first printed prototype, modelled by the team in Onshape, a conventional CAD package.

## Author contributions

S.C.: conceptualization, methodology, investigation, validation, data curation, writing – original draft, writing – review & editing. W.M.: methodology, software, investigation, formal analysis, validation. L.W.: software, investigation, resources. C.R.: investigation. G.E.: investigation. S.G.B.: conceptualization, methodology, software, formal analysis, visualization, resources, supervision, project administration, funding acquisition, writing – original draft, writing – review & editing.

## Conflicts of interest

There are no conflicts to declare.

## Data availability

Data for this article, including the raw test-protocol records, the per-dose record, the tidy datasets behind Figs. 3–5, the 128-entry design log, and the AI prompt and response logs, are available in the powder-doser repository at `https://github.com/vertical-cloud-lab/powder-doser` and in its archived release on Zenodo.[44] The code for the doser firmware (MicroPython), the analysis and figure scripts (Python), and the parametric CAD source (CadQuery, OpenSCAD, and KCL) can be found at the same repository and release. The version of the code employed for this study is version 1.0.0. The repository also holds the team's Fusion 360 parts as STEP and STL files, renders, an animated assembly guide, and the KiCad schematics. The assembled current design is a public Onshape document (`https://cad.onshape.com/documents/ae9f107d3972fc9d390e541f`), and each printed part has a public Fusion 360 link (SI Table S2). Running `paper/figures/make_data_figures.py` rebuilds the data figures, and the scripts in `paper/figures/data/` rebuild the SI tables. Hardware designs, code, and data are released under the open licenses stated in the repository. The SI summarizes every run and every dose attempt in tables. References cited in the SI are listed in this article's reference list.

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We thank Devora Najjar and Ron Darling, who started this project with one of us (S.G.B.) at a two-day hackathon in April 2026. Devora Najjar wrote the first auger model there, and Nasa Sinnott-Armstrong helped print it. We thank Philip Lampkin and Jacob Lessard (Department of Chemistry, University of Utah) for hosting the glove-box trial and for their advice on dosing in synthetic chemistry. Generative AI tools were used extensively and deliberately in this work, both as objects of study and as assistants (SI Table S12). During the preparation of this study and manuscript, the authors used the following tools. Three LLM tools wrote CAD code, firmware, and analysis and figure scripts, and drafted and revised the text of the manuscript and SI: the GitHub Copilot coding agent (GitHub; mostly Claude Opus 4.7 and Opus 4.8, April to July 2026), a local Claude session (Claude Sonnet 4.6, April 2026), and Claude Code (Anthropic; Claude Opus 4.8, Opus 5, Opus 5.5, Fable 5, and Fable 5.1, June to October 2026). Edison Scientific searched the literature, compiled references, and ran mock reviews of designs, data, and drafts. CADSmith, the Zoo Text-to-CAD API, and Zoo Design Studio (through its Zookeeper agent) served as text-to-CAD tools, and GitHub Copilot code review reviewed pull requests automatically. The authors have reviewed and edited the output and take full responsibility for the content of this publication.

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